NVIDIA Gives OmniVinci: Model Can See, Read, Listen, Speak, Reason - Run Locally

Fahd Mirza
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408 views 2 hours ago #omnivinci
This video locally installs OmniVinci, an omni-modal LLM that can not only see and read but also listen, speak, and reason.

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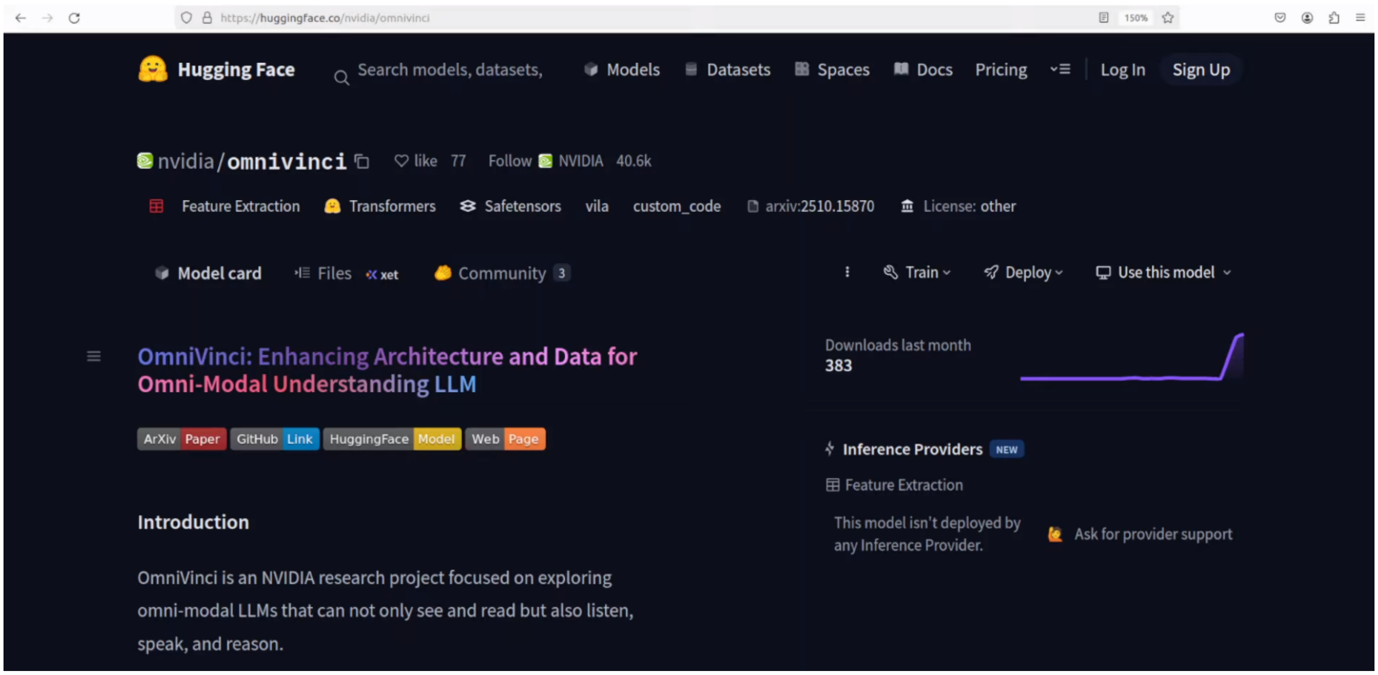
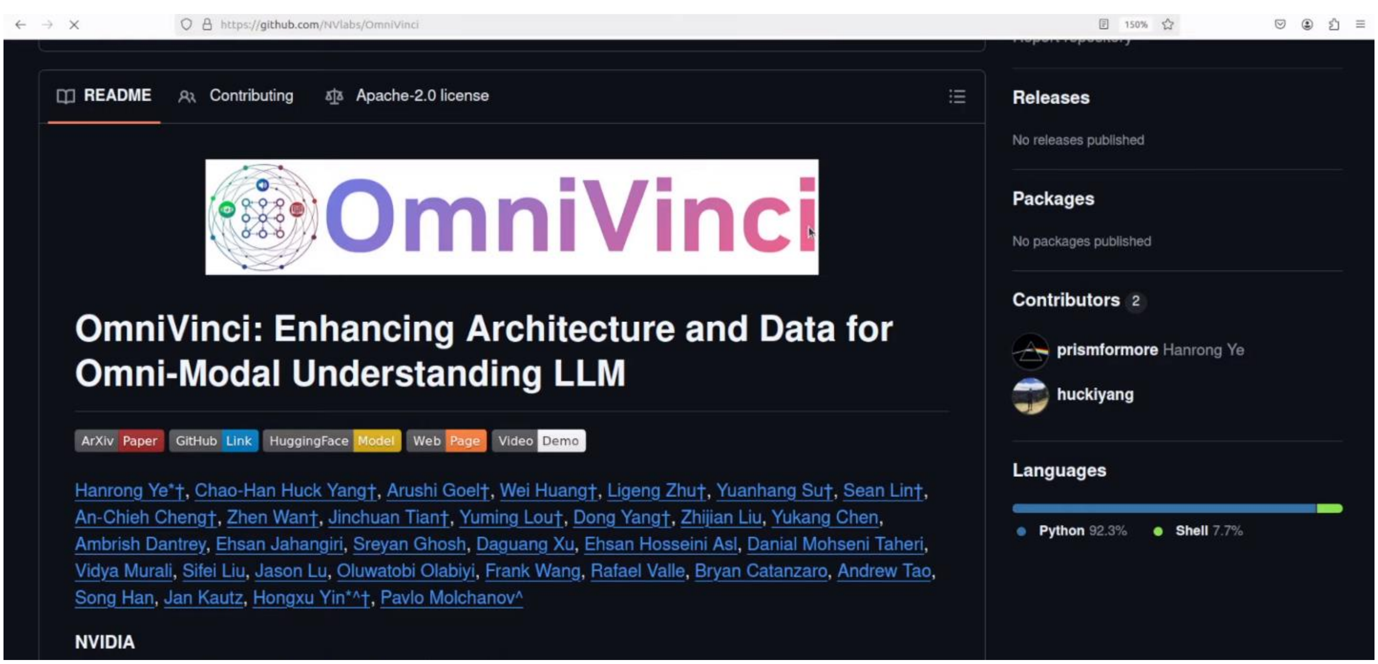
AI-generated video summary · Quality and accuracy may vary.

See, read, listen, speak, and reason – all locally! This video demonstrates installing NVIDIA's OmniVinci, an omni-modal large language model. The tutorial explores its architecture, showcasing its capabilities with diverse video examples.





```
Ubuntu@0151-dsm-prxmx30181:~$ git clone https://github.com/NVlabs/OmniVinci.git && cd OmniVinci
Cloning into 'OmniVinci'...
remote: Enumerating objects: 133, done.
remote: Counting objects: 100% (61/61), done.
remote: Compressing objects: 100% (41/41), done.
remote: Total 133 (delta 35), reused 36 (delta 20), pack-reused 72 (from 1)
Receiving objects: 100% (133/133), 99.01 MiB | 45.22 MiB/s, done.
Resolving deltas: 100% (52/52), done.
Ubuntu@0151-dsm-prxmx30181:~/OmniVinci$ pip install huggingface_hub
```

```
hub
  WARNING: The script tqdm is installed in '/home/Ubuntu/.local/bin' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script
-location.
  WARNING: The scripts hf, huggingface-cli and tiny-agents are installed in '/home/Ubuntu/.local/bin' whi
ch is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script
-location.
Successfully installed filelock-3.20.0 fsspec-2025.9.0 hf-xet-1.2.0 huggingface_hub-0.36.0 packaging-25.0
 tqdm-4.67.1 typing-extensions-4.15.0
Ubuntu@0151-dsm-prxmx30181:~/OmniVinci$
```

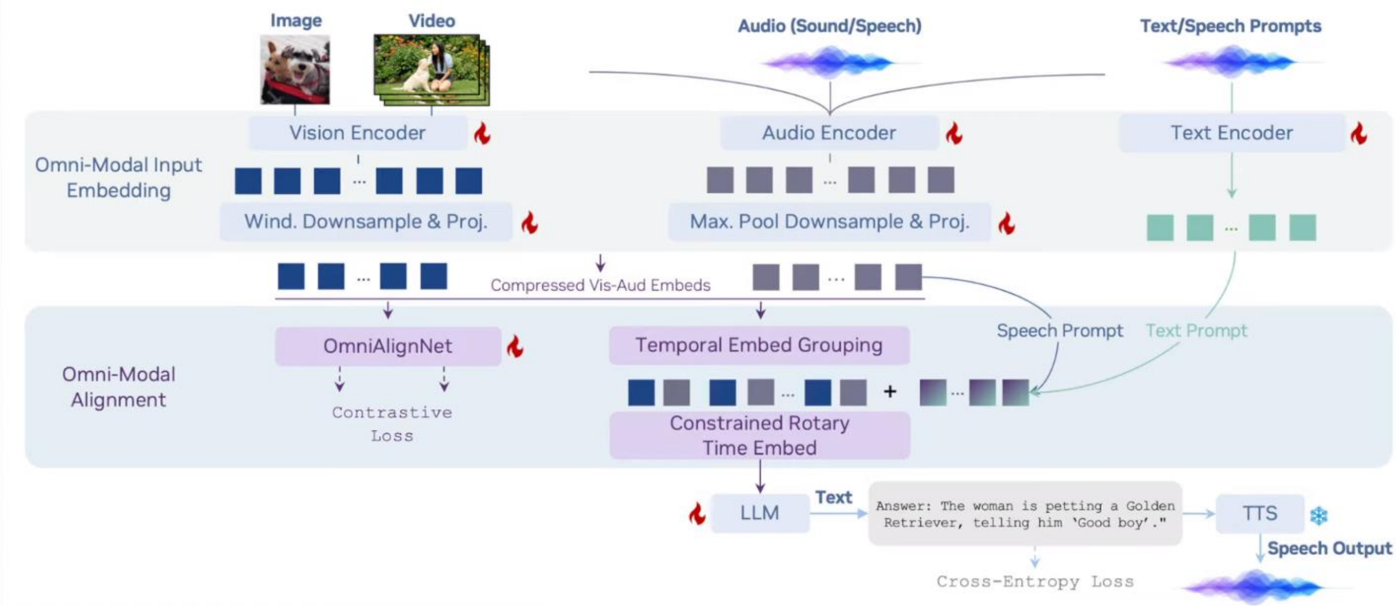
```
Ubuntu@0151-dsm-prxmx30181: ~/OmniVinci
Ubuntu@0151-dsm-prxmx30181:~/OmniVinci$ hf auth login
hf: command not found
Ubuntu@0151-dsm-prxmx30181:~/OmniVinci$
```

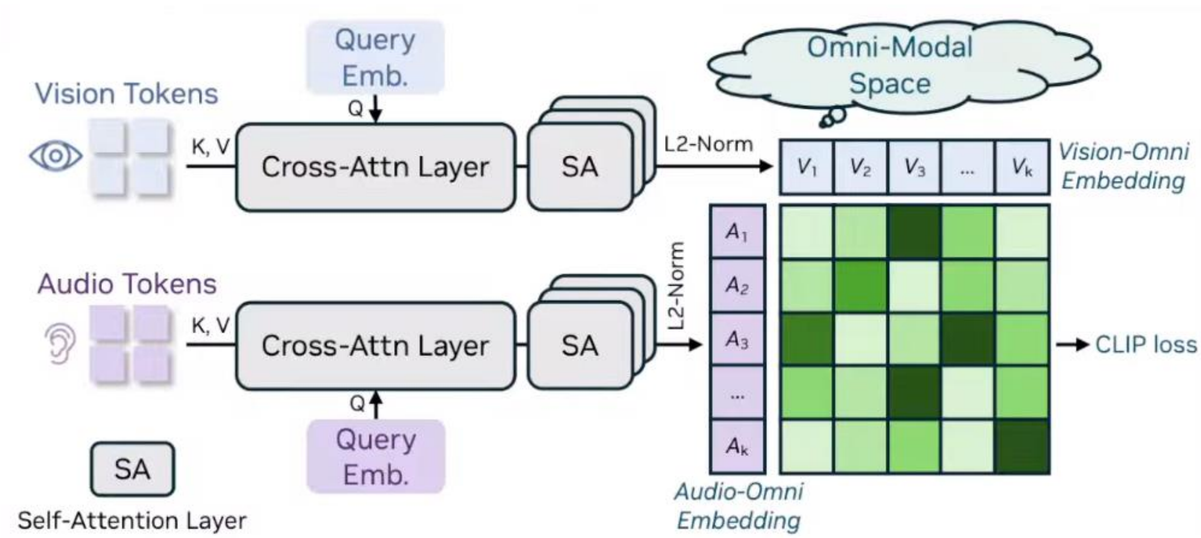
```
Ubuntu@0151-dsm-prxmx30181: ~/OmniVinci
Ubuntu@0151-dsm-prxmx30181:~/OmniVinci$ huggingface-cli download nvidia/omnivinci --local-dir ./omnivinci
 --local-dir-use-symlinks False
cd ./omnivinci
/home/Ubuntu/.local/lib/python3.10/site-packages/huggingface_hub/commands/download.py:141: FutureWarning:
 Ignoring --local-dir-use-symlinks. Downloading to a local directory does not use symlinks anymore.
  warnings.warn(
⚠ Warning: 'huggingface-cli download' is deprecated. Use 'hf download' instead.
Fetching 57 files:   0%|                                                  | 0/57 [00:00<?, ?it/s]
```

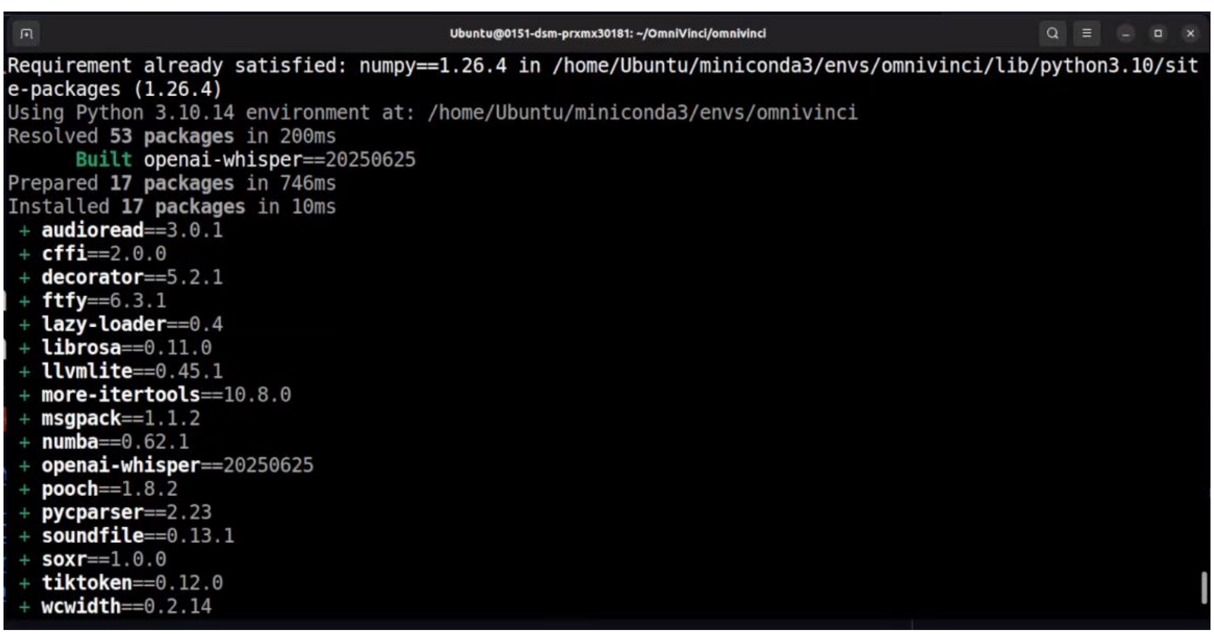
```
llm/model-00002-of-00004.safetensors: 100%|                    | 4.93G/4.93G [00:23<00:00, 207MB/s]
Download complete. Moving file to omnivinci/llm/model-00002-of-00004.safetensors7G [00:23<00:00, 284MB/s]
llm/model-00001-of-00004.safetensors: 100%|                    | 4.87G/4.87G [00:24<00:00, 203MB/s]
Download complete. Moving file to omnivinci/llm/model-00001-of-00004.safetensors7G [00:24<00:00, 225MB/s]
llm/model-00003-of-00004.safetensors: 100%|                    | 4.33G/4.33G [00:24<00:00, 180MB/s]
Download complete. Moving file to omnivinci/llm/model-00003-of-00004.safetensors3G [00:23<00:00, 603MB/s]
Fetching 57 files: 100%|                                       | 57/57 [00:25<00:00,  2.23it/s]
/home/Ubuntu/OmniVinci/omnivincisors: 100%|                    | 4.33G/4.33G [00:24<00:00, 713MB/s]
Ubuntu@0151-dsm-prxmx30181:~/OmniVinci/omnivinci$
```

```
Ubuntu@0151-dsm-prxmx30181: ~/OmniVinci/omnivinci
Ubuntu@0151-dsm-prxmx30181:~/OmniVinci/omnivinci$ bash ./environment_setup.sh omnivinci
Channels:
 - defaults
Platform: linux-64
Collecting package metadata (repodata.json): -
```

```
libzlib-1.3.1        | 59 KB     | ###############7                                          |  27%
_openmp_mutex-5.1    | 21 KB     | #############################################6            |  77%
xorg-libxdmcp-1.1.5  | 19 KB     | ##################################################7       |  85%
xorg-libxau-1.0.12   | 13 KB     | ########################################################## | 100%
pthread-stubs-0.3    | 5 KB      | ########################################################## | 100%
_libgcc_mutex-0.1    | 3 KB      | ########################################################## | 100%
```








```
Requirement already satisfied: numpy==1.26.4 in /home/Ubuntu/miniconda3/envs/omnivinci/lib/python3.10/sit
e-packages (1.26.4)
Using Python 3.10.14 environment at: /home/Ubuntu/miniconda3/envs/omnivinci
Resolved 53 packages in 200ms
      Built openai-whisper==20250625
Prepared 17 packages in 746ms
Installed 17 packages in 10ms
 + audioread==3.0.1
 + cffi==2.0.0
 + decorator==5.2.1
 + ftfy==6.3.1
 + lazy-loader==0.4
 + librosa==0.11.0
 + llvmlite==0.45.1
 + more-itertools==10.8.0
 + msgpack==1.1.2
 + numba==0.62.1
 + openai-whisper==20250625
 + pooch==1.8.2
 + pycparser==2.23
 + soundfile==0.13.1
 + soxr==1.0.0
 + tiktoken==0.12.0
 + wcwidth==0.2.14
```

Everything is now installed. Next, let us use the script below to test the model using a multimodal (video + voice) prompt

File Edit Selection View Go Run Terminal Help

EXPLORER

MYCODE

app.py

```python
from transformers import AutoProcessor, AutoModel, AutoConfig,AutoModelForCausalLM
import torch
import os

# default: Load the model on the available device(s)
model_path = "./"
video_path = "/home/Ubuntu/videos/fahd.mp4"
generation_kwargs = {"max_new_tokens": 1024, "max_length": 99999999}
load_audio_in_video = True
num_video_frames = 128
audio_length = "max_3600"

config = AutoConfig.from_pretrained(model_path, trust_remote_code=True)

model = AutoModel.from_pretrained(model_path,
                                  trust_remote_code=True,
                                  torch_dtype="torch.float16",
                                  device_map="auto")

```

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EXPLORER

MYCODE

app.py

```python

config = AutoConfig.from_pretrained(model_path, trust_remote_code=True)

model = AutoModel.from_pretrained(model_path,
                                  trust_remote_code=True,
                                  torch_dtype="torch.float16",
                                  device_map="auto")

processor = AutoProcessor.from_pretrained(model_path, trust_remote_code=True)
generation_config = model.default_generation_config
generation_config.update(**generation_kwargs)

model.config.load_audio_in_video = load_audio_in_video
processor.config.load_audio_in_video = load_audio_in_video
if num_video_frames > 0:
    model.config.num_video_frames = num_video_frames
    processor.config.num_video_frames = num_video_frames
if audio_length != -1:
    model.config.audio_chunk_length = audio_length
    processor.config.audio_chunk_length = audio_length


conversation = [{
        "role": "user",
        "content": [
            {"type": "video", "video":video_path},
            {"type": "text", "text": "Assess the video, followed by a detailed description of it's
        ]
}]
```

OUTLINE

TIMELINE

File Edit Selection View Go Run Terminal Help

EXPLORER

MYCODE

app.py

```python
generation_config = model.default_generation_config
generation_config.update(**generation_kwargs)

model.config.load_audio_in_video = load_audio_in_video
processor.config.load_audio_in_video = load_audio_in_video
if num_video_frames > 0:
    model.config.num_video_frames = num_video_frames
    processor.config.num_video_frames = num_video_frames
if audio_length != -1:
    model.config.audio_chunk_length = audio_length
    processor.config.audio_chunk_length = audio_length


conversation = [{
        "role": "user",
        "content": [
            {"type": "video", "video":video_path},
            {"type": "text", "text": "Assess the video, followed by a detailed description of it's \
        ]
}]
text = processor.apply_chat_template(conversation, tokenize=False, add_generation_prompt=True)

inputs = processor([text])

output_ids = model.generate(
    input_ids=inputs.input_ids,
    media=getattr(inputs, 'media', None),
    media_config=getattr(inputs, 'media_config', None),
    generation_config=generation_config,
```

OUTLINE

TIMELINE

```python
video":video_path},
ext": "Assess the video, followed by a detailed description of it's video and audio contents."}


plate(conversation, tokenize=False, add_generation_prompt=True)
```

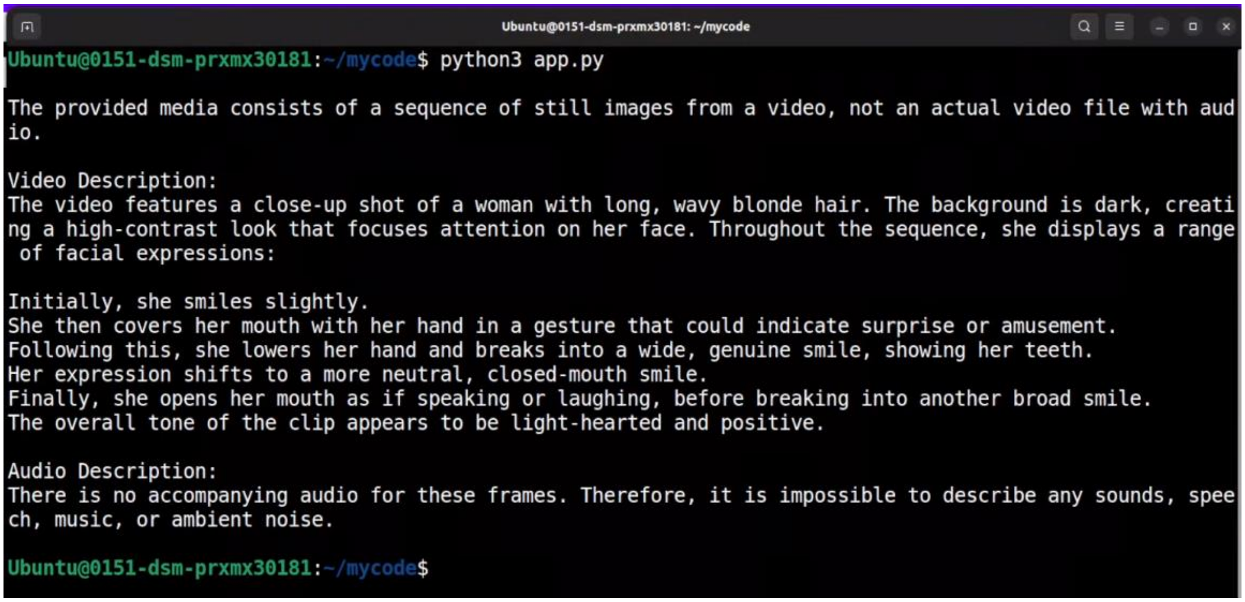
Ubuntu@0151-dsm-prxmx30181: ~/mycode

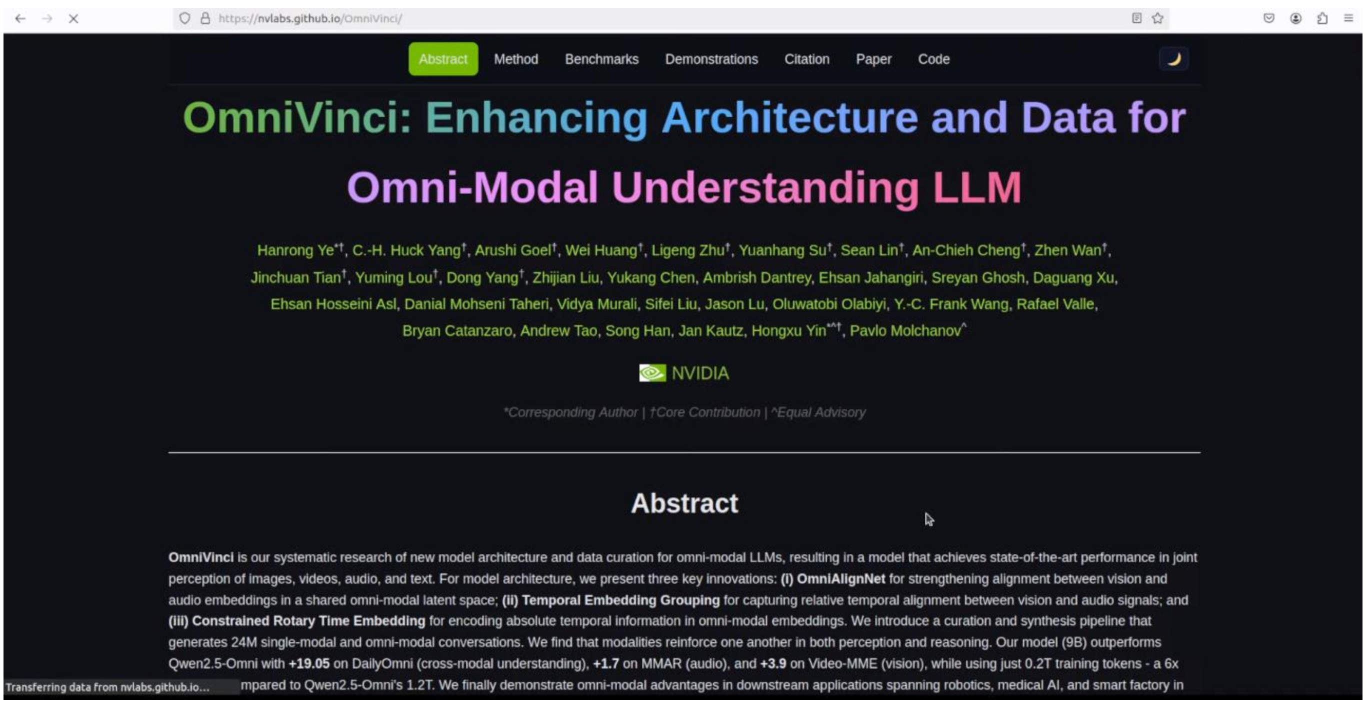
```
Ubuntu@0151-dsm-prxmx30181:~/mycode$ python app.py
```

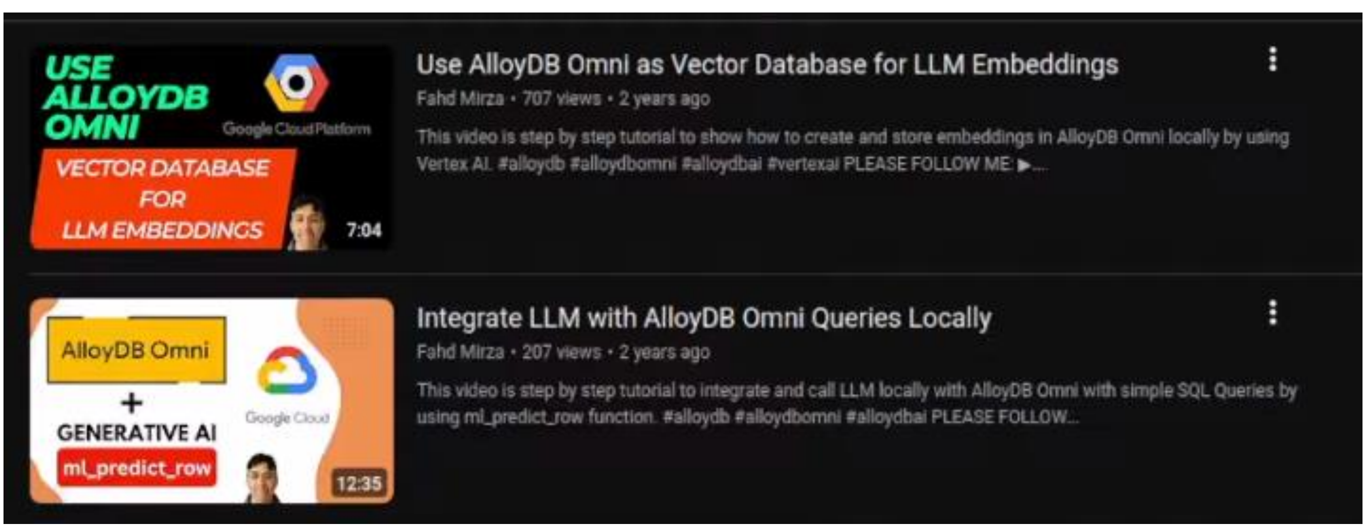
```
Ubuntu@0151-dsm-prxmx30181: ~
Ubuntu@0151-dsm-prxmx30181:~$ nvtop
```

```
    PID  USER DEV    TYPE  GPU        GPU MEM    CPU  HOST MEM Command
  12886 Ubuntu   0 Compute   0%  72204MiB  89%     0%    487MiB python3 gpu.py 70 --device 0 --reserve-mb 1024
   1342   root   0 Graphic   0%    108MiB   0%     0%     68MiB /usr/lib/xorg/Xorg vt1 -displayfd 3 -auth /run/u
   2362    gdm   0 Graphic   0%      9MiB   0%     0%    223MiB /usr/bin/gnome-shell

F2Setup  F6Sort  F9Kill  F10Quit  F12Save Config
```

```
Ubuntu@0151-dsm-prxmx30181: ~/mycode
Ubuntu@0151-dsm-prxmx30181:~/mycode$ python3 app.py

Video Assessment:
The video features a man sitting at a desk and speaking directly to the camera. It appears to be a casual
, self-recorded video, likely for a social media platform or a personal blog. The man is demonstrating or
 testing a "deep live cam" and mentions creating a "sample short video" for this purpose. He clarifies th
at the video is for educational purposes and asks viewers not to judge his casual attire as it is a Satur
day morning. He concludes by encouraging viewers to subscribe to his channel and share the content. The o
verall tone is informal and friendly.
Video and Audio Description:
Video: A middle-aged man with dark, medium-length hair pulled back, and some graying hair, is shown from
the chest up. He is wearing a dark hooded sweatshirt. He is seated in a chair in what appears to be a hom
e office. Behind him, to the right, is a white desk with a laptop on a stand, an external monitor, a keyb
oard, and a mouse. The window behind the desk has brown blinds that are closed. The wall behind him is a
light, neutral color. The man gestures with his hands as he speaks.
Audio: The man speaks in a clear but somewhat quiet voice. He explains that he is creating a short video
as a demo for a "deep live cam." He states the video is for educational purposes and mentions it is Satur
day morning in Sydney. He asks viewers to subscribe to his channel and share it with their network before
 thanking them for watching. There are no other discernible background noises.

Ubuntu@0151-dsm-prxmx30181:~/mycode$
```


Ubuntu@0151-dsm-prxmx30181: ~/mycode
Ubuntu@0151-dsm-prxmx30181:~/mycode$ python3 app.py
The provided media consists of a sequence of still images from a video, not an actual video file with audio.
Video Description:
The video features a close-up shot of a woman with long, wavy blonde hair. The background is dark, creating a high-contrast look that focuses attention on her face. Throughout the sequence, she displays a range of facial expressions:
Initially, she smiles slightly.
She then covers her mouth with her hand in a gesture that could indicate surprise or amusement.
Following this, she lowers her hand and breaks into a wide, genuine smile, showing her teeth.
Her expression shifts to a more neutral, closed-mouth smile.
Finally, she opens her mouth as if speaking or laughing, before breaking into another broad smile.
The overall tone of the clip appears to be light-hearted and positive.
Audio Description:
There is no accompanying audio for these frames. Therefore, it is impossible to describe any sounds, speech, music, or ambient noise.
Ubuntu@0151-dsm-prxmx30181:~/mycode$


https://nvlabs.github.io/OmniVinci/
Abstract
Method
Benchmarks
Demonstrations
Citation
Paper
Code
OmniVinci: Enhancing Architecture and Data for Omni-Modal Understanding LLM
Hanrong Ye*†, C.-H. Huck Yang†, Arushi Goel†, Wei Huang†, Ligeng Zhu†, Yuanhang Su†, Sean Lin†, An-Chieh Cheng†, Zhen Wan†, Jinchuan Tian†, Yuming Lou†, Dong Yang†, Zhijian Liu, Yukang Chen, Ambrish Dantrey, Ehsan Jahangiri, Sreyan Ghosh, Daguang Xu, Ehsan Hosseini Asl, Danial Mohseni Taheri, Vidya Murali, Sifei Liu, Jason Lu, Oluwatobi Olabiyi, Y.-C. Frank Wang, Rafael Valle, Bryan Catanzaro, Andrew Tao, Song Han, Jan Kautz, Hongxu Yin*^†, Pavlo Molchanov^
NVIDIA
*Corresponding Author | †Core Contribution | ^Equal Advisory
Abstract
OmniVinci is our systematic research of new model architecture and data curation for omni-modal LLMs, resulting in a model that achieves state-of-the-art performance in joint perception of images, videos, audio, and text. For model architecture, we present three key innovations: (i) OmniAlignNet for strengthening alignment between vision and audio embeddings in a shared omni-modal latent space; (ii) Temporal Embedding Grouping for capturing relative temporal alignment between vision and audio signals; and (iii) Constrained Rotary Time Embedding for encoding absolute temporal information in omni-modal embeddings. We introduce a curation and synthesis pipeline that generates 24M single-modal and omni-modal conversations. We find that modalities reinforce one another in both perception and reasoning. Our model (9B) outperforms Qwen2.5-Omni with +19.05 on DailyOmni (cross-modal understanding), +1.7 on MMAR (audio), and +3.9 on Video-MME (vision), while using just 0.2T training tokens - a 6x
Transferring data from nvlabs.github.io...
mpared to Qwen2.5-Omni's 1.2T. We finally demonstrate omni-modal advantages in downstream applications spanning robotics, medical AI, and smart factory in


USE ALLOYDB OMNI
Google Cloud Platform
VECTOR DATABASE FOR LLM EMBEDDINGS
7:04
Use AlloyDB Omni as Vector Database for LLM Embeddings
Fahd Mirza • 707 views • 2 years ago
This video is step by step tutorial to show how to create and store embeddings in AlloyDB Omni locally by using Vertex AI. #alloydb #alloydbomni #alloydbai #vertexai PLEASE FOLLOW ME: ▶...
AlloyDB Omni
+
GENERATIVE AI
ml_predict_row
Google Cloud
12:35
Integrate LLM with AlloyDB Omni Queries Locally
Fahd Mirza • 207 views • 2 years ago
This video is step by step tutorial to integrate and call LLM locally with AlloyDB Omni with simple SQL Queries by using ml_predict_row function. #alloydb #alloydbomni #alloydbai PLEASE FOLLOW...